



UNIMORE

UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

Dipartimento di Ingegneria
“Enzo Ferrari”

Fondamenti di Elettronica Digitale

Power Supply

Laurea Triennale in Ingegneria Informatica

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Non-Linear Circuits

DC-DC Converters

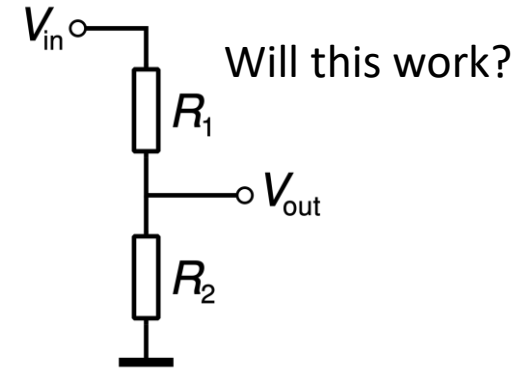
Up to now we have seen many circuits (and we are going to see more) but we neglected an important aspect: supply!

All active circuits must be supplied, and they need appropriate bias voltages. The latter could be different in different circuits and/or sub-circuits in your system and must be properly generated.

The vehicle is biased, overall, by a battery.

- It only offers one bias voltage (i.e., 12 V)
 - Possible solution could be using many batteries but it would be expensive from many points of view
- The generated voltage is not constant over time
 - A general issue that requires circuitry that is somehow insensitive to its bias voltage to be used to generate a constant voltage over time

$$V_{out} = V_{in} \frac{R_2 || R_{LOAD}}{R_1 + R_2 || R_{LOAD}}$$



We will have a look at:

- Linear regulators
 - Linear circuits that provide a step-down voltage conversion. Very accurate and low noise.
- Switching converters
 - Circuits that use the concept of switching to realize both step-up and step-down conversion. Less accurate and more noisy.

Non-Linear Circuits

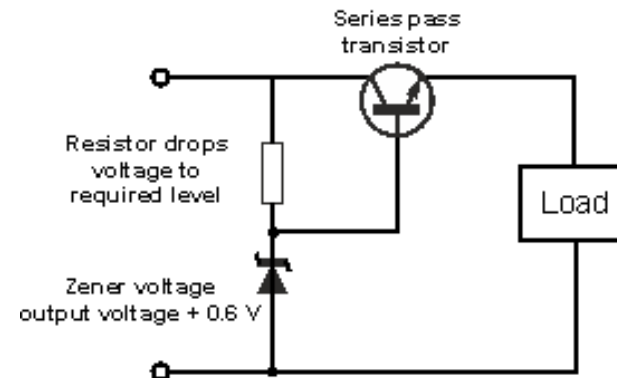
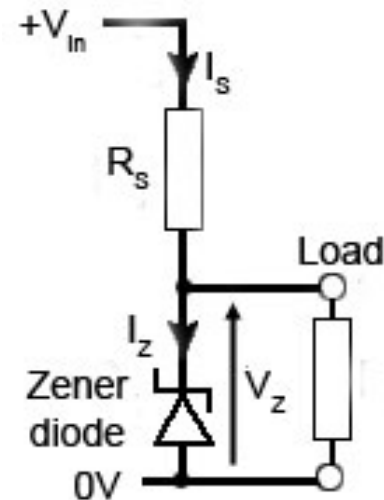
Linear Regulators

It acts as a controlled resistance.

The regulating device is made to act like a variable resistor, continuously adjusting a voltage divider network to maintain a constant output voltage and continually dissipating the difference between the input and regulated voltages as **waste heat**.

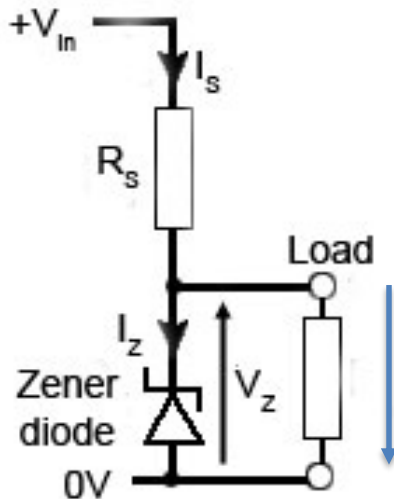
The disadvantage is that, since the regulated voltage of a linear regulator must always be lower than input voltage, efficiency is limited and the input voltage must be high enough to always allow the active device to drop some voltage. **Power is wasted!**

Shunt and series regulators. Series reg. has better regulation than the simple shunt reg., since the base current of the transistor forms a very light load on the Zener, thereby minimizing variation in Zener voltage due to variation in the load.



Non-Linear Circuits

Linear Regulators



$$V_{zener} = 5V$$

$$I_{soglia} = 1mA$$

$$R_s = 1k\Omega$$

$$I_{max} = 20mA$$

$$R_{load} = 5k\Omega$$

$$I_s = I_z + I_l = I_z + \frac{V_z}{R_{load}} = I_z + 1mA$$

$$I_{smin} = I_{zmin} + \frac{V_z}{R_{load}} = 1mA + 1mA = 2mA$$

$$V_{rsmin} = I_{smin} * R_s = 2V$$

$$I_{smax} = I_{zmax} + \frac{V_z}{R_{load}} = 20mA + 1mA = 21mA$$

$$V_{rsmax} = I_{smax} * R_s = 21V$$

$$V_{inmin} = 7V$$

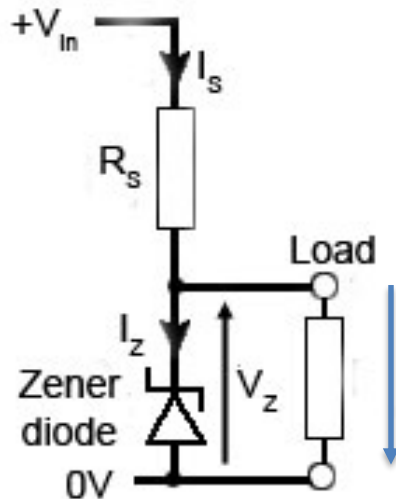
$$V_{inmax} = 26V$$

$$P_{in} = 7V * 2mA = 14mW$$

$$P_{load} = 5V * 1mA = 5mW \quad \eta = \frac{P_{load}}{P_{in}} = \frac{5}{14} = 35.7\%$$

Non-Linear Circuits

Linear Regulators



$$V_{zener} = 5V$$

$$I_{soglia} = 1mA$$

$$R_s = 1k\Omega$$

$$I_{max} = 20mA$$

$$V_{in} = 30V$$

$$I_s = 25mA$$

$$I_{lmax} = I_s - I_{zmin} = 25mA - 1mA = 24mA$$

$$R_{lmin} = \frac{V_{zener}}{I_{lmax}} = 208\Omega$$

$$I_{lmin} = I_s - I_{zmax} = 25mA - 20mA = 5mA$$

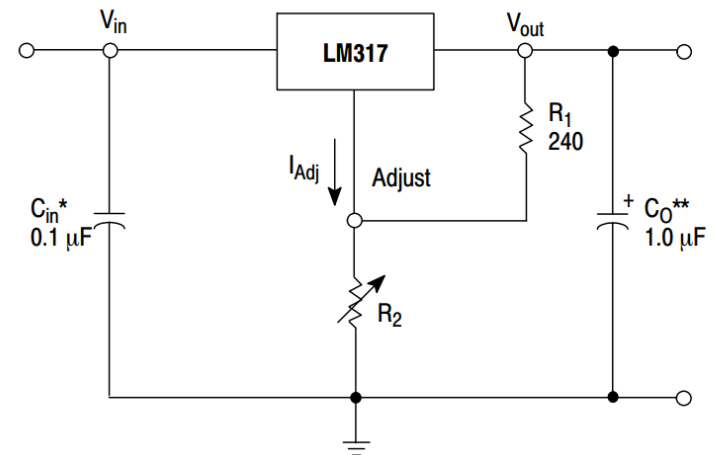
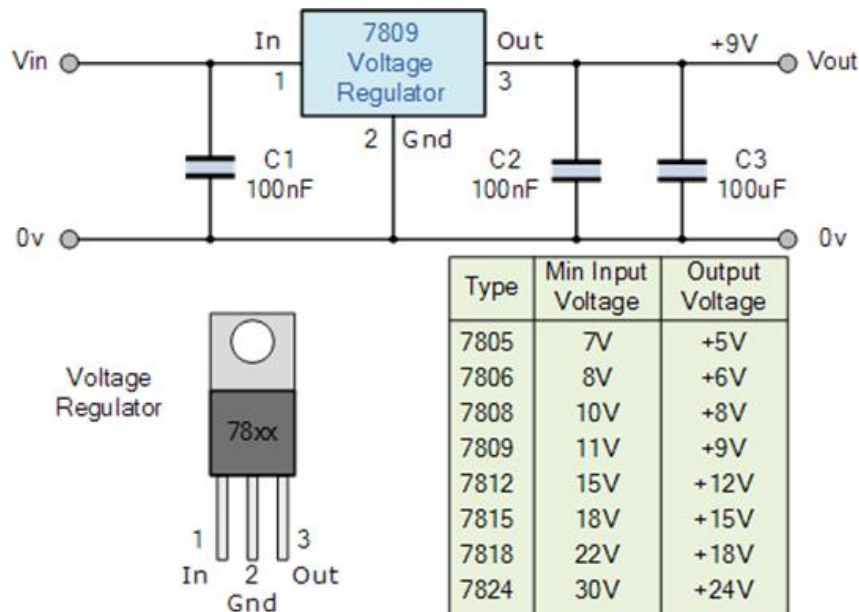
$$R_{lmax} = \frac{V_{zener}}{I_{lmin}} = 1k\Omega$$

Non-Linear Circuits

Linear Regulators

Fixed regulators are typically available in many flavors (i.e., many possible output voltages). Available as 3-terminal devices. An example is the 78xx and 79xx series.

There are also Adjustable Regulators (e.g., LM317) and Low Dropout Regulators.



* C_{in} is required if regulator is located an appreciable distance from power supply filter.

** C_O is not needed for stability, however, it does improve transient response.

$$V_{out} = 1.25 V \left(1 + \frac{R_2}{R_1} \right) + I_{Adj} R_2$$

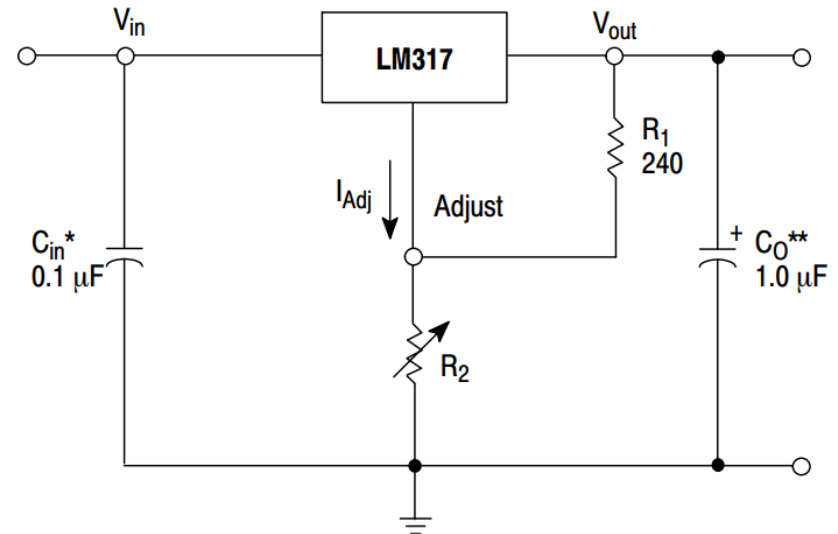
Since I_{Adj} is controlled to less than $100 \mu A$, the error associated with this term is negligible in most applications.

Non-Linear Circuits

Linear Regulators

In operation, the LM317 develops and maintains a nominal 1.25 V reference between its output and adjustment terminals. This reference voltage is converted to a programming current by R_1 , and this constant current flows through R_2 to ground.

Since the current from the adjustment terminal (I_{Adj}) represents an error term in the equation, the LM317 was designed to control it to less than 100 μA and keep it constant. Since the LM317 is a floating regulator, it is only the voltage differential across the circuit which is important to performance, and operation at high voltages with respect to ground is possible.



* C_{in} is required if regulator is located an appreciable distance from power supply filter.

** C_O is not needed for stability, however, it does improve transient response.

$$V_{out} = 1.25 \text{ V} \left(1 + \frac{R_2}{R_1} \right) + I_{Adj} R_2$$

Since I_{Adj} is controlled to less than 100 μA , the error associated with this term is negligible in most applications.

Non-Linear Circuits

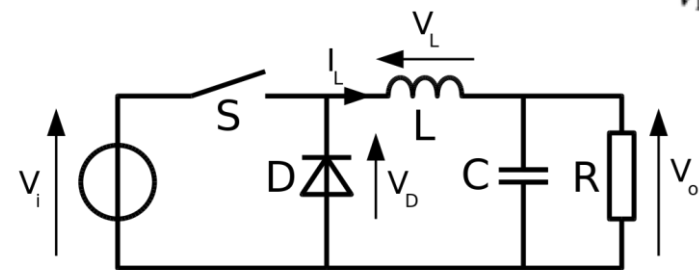
DC-DC Converters - Buck

- Circuits that use the concept of switching to realize both step-up and step-down conversion.
- Less accurate and more noisy. Uses bulky inductor.
- Very efficient (typically $\eta > 90\%$)

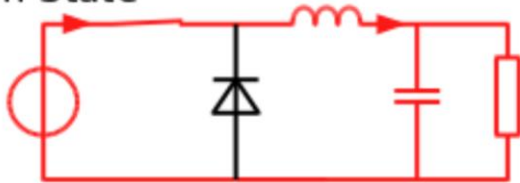
$$V_L = L \frac{dI_L}{dt}$$

$$\Delta I_{L_{on}} = \int_0^{t_{on}} \frac{V_L}{L} dt = \frac{(V_i - V_o)}{L} t_{on}, t_{on} = DT$$

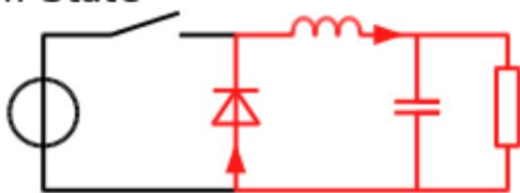
$$\Delta I_{L_{off}} = \int_{t_{on}}^{T=t_{on}+t_{off}} \frac{V_L}{L} dt = -\frac{V_o}{L} t_{off}, t_{off} = (1 - D)T$$



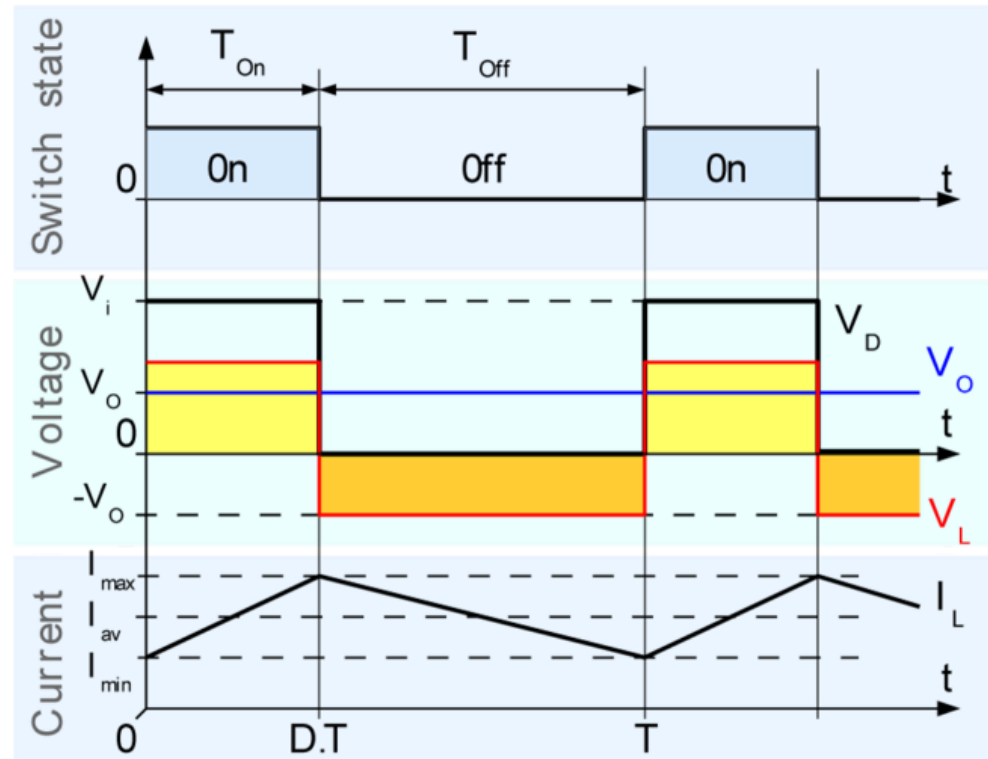
On-State



Off-State



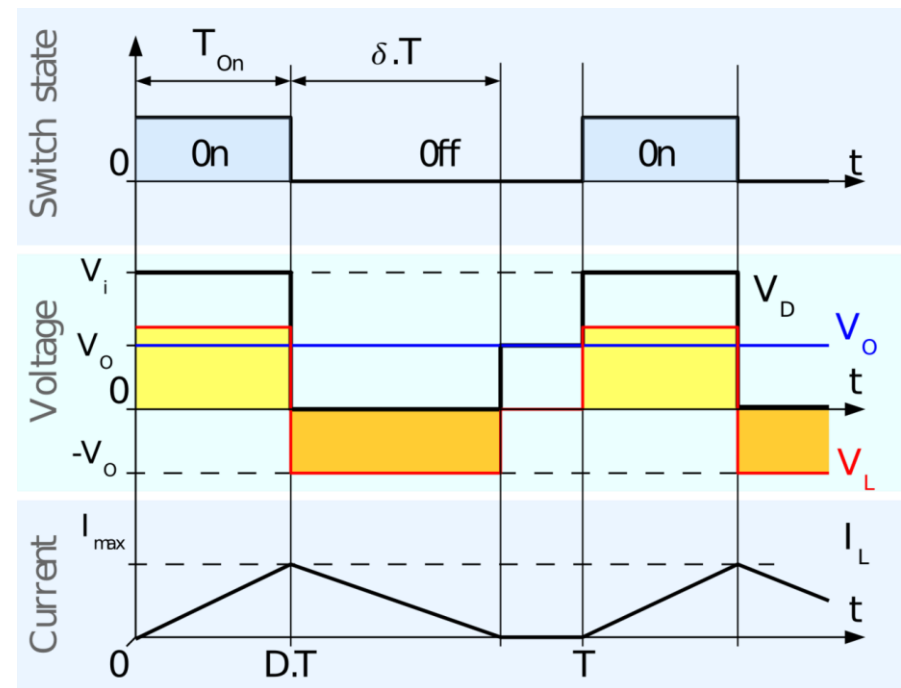
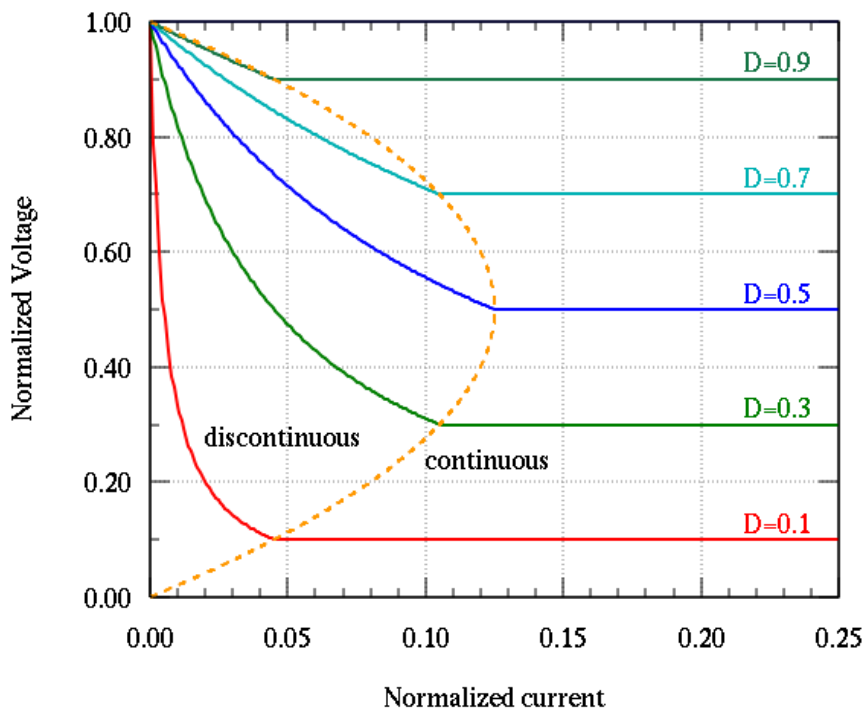
$$D = \frac{V_o}{V_i}$$



Non-Linear Circuits

DC-DC Converters - Buck

- Can go in discontinuous mode (DCM) if the load is very light (i.e., demands low energy or current)
- In DCM the inductor current goes to zero during the switching cycle
- V_O depends now on almost all circuit parameters (sadly, also I_O , hence the load) instead of only on D
- A benefit of the DCM is that the inductor could be made much smaller
- Another benefit is that the on commutation of the switch happens at 0 current, minimizing the losses

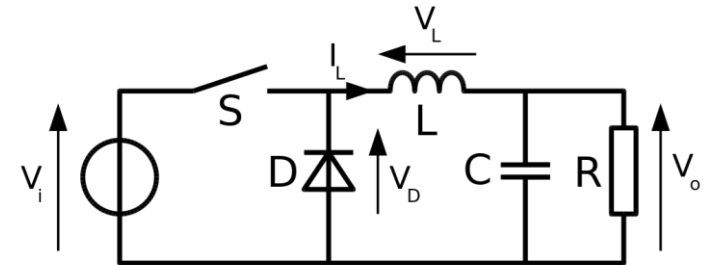


Non-Linear Circuits

DC-DC Converters - Buck

Losses are mainly related to:

- Diode voltage drop during the off period
 - Reduced by using Schottky diode or Synchronous buck
- The on resistance of the switch transistor
 - Low R_{DSon} switch required
- The transition from off to on and vice-versa
 - Depends on f_{sw} . Higher f_{sw} means more transitions through the high dissipation region of the switch. Above a certain f_{sw} , these type of losses become dominant. Difficult to calculate accurately without simulations.
- The series resistance of the capacitor
- The series resistance of the inductor

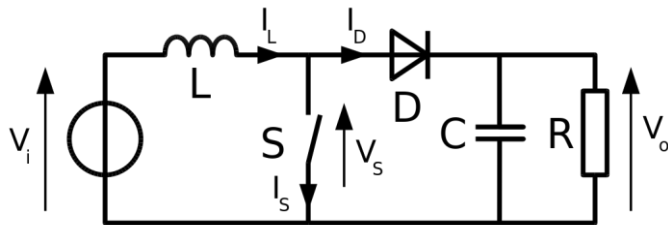


Cause	Power Lost
Diode	$P_D = V_D I_{L(off)} (1 - D)$
Switch on resistance	$P_{on} = I_{L(on)(rms)}^2 R_{on} D$
Inductor DCR	$P_L = \left(I_{out}^2 + \frac{\Delta I_L^2}{12} \right) R_L$
Input Capacitor ESR	$P_{C_{in}} = D \left(I_{out}^2 (1 - D) + \frac{\Delta I_L^2}{12} \right) R_C$
Output Capacitor ESR	$P_{C_{out}} = \frac{\Delta I_L^2}{12} R_C$
Transistor switching	$P_{SW} + P_G = (V I_{L(on)} (t_{rise} + t_{fall}) + Q_G V_{GS}) f_{sw}$

Non-Linear Circuits

DC-DC Converters - Boost

- Circuits that use the concept of switching to realize both step-up and step-down conversion.
- Less accurate and more noisy. Uses bulky inductor.
- Very efficient (typically $\eta > 90\%$)

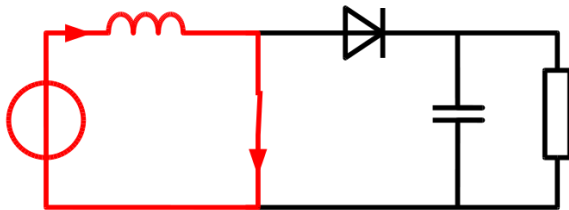


$$V_L = L \frac{dI_L}{dt}$$

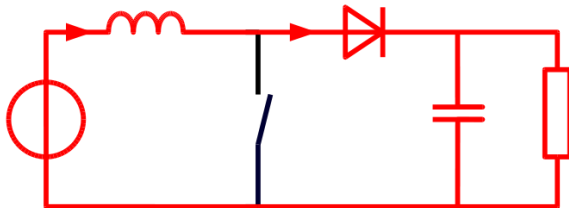
$$\Delta I_{L_{On}} = \frac{1}{L} \int_0^{DT} V_i dt = \frac{DT}{L} V_i$$

$$\Delta I_{L_{Off}} = \int_{DT}^T \frac{(V_i - V_o) dt}{L} = \frac{(V_i - V_o)(1 - D)T}{L}$$

On-State



Off-State



$$\frac{V_o}{V_i} = \frac{1}{1 - D}$$

